Модуль 2

```
CREATE TABLE roles (
roles_id SERIAL PRIMARY KEY,
roles_name VARCHAR(50)
);

CREATE TABLE users (
users_id SERIAL PRIMARY KEY,
login VARCHAR(100),
last_name VARCHAR(100),
first_name VARCHAR(100),
middle_name VARCHAR(100),
roles_id INTEGER REFERENCES roles (roles_id)
);

CREATE TABLE category (
category_id SERIAL PRIMARY KEY,
category_name VARCHAR(100)
);

CREATE TABLE sizes(
sizes_id SERIAL PRIMARY KEY,
sizes_name VARCHAR(100)
);

CREATE TABLE model(
model_id SERIAL PRIMARY KEY,
model_name VARCHAR(100),
manufacture VARCHAR(100),
image_url VARCHAR(100),
description TEXT,
composition TEXT,
price NUMERIC(10, 2) NOT NULL,
category_id INT REFERENCES category (category_id)
);

CREATE TABLE product (
product_id SERIAL PRIMARY KEY,
model_id INT REFERENCES model (model_id),
sizes_id INT REFERENCES sizes (sizes_id),
product_name VARCHAR(100)
);
```

```sql
CREATE TABLE orders (
orders_id SERIAL PRIMARY KEY,
users_id INT REFERENCES users (users_id),
order_date DATE NOT NULL DEFAULT CURRENT_DATE,
status VARCHAR(20) NOT NULL DEFAULT 'new'
);

CREATE TABLE orders_item (
orders_item_id SERIAL PRIMARY KEY,
orders_id INT REFERENCES orders (orders_id),
product_id INT REFERENCES product (product_id),
model_id INT REFERENCES model (model_id),
quantity INT CHECK (quantity > 0)
);
```

```python
import tkinter as tk
from tkinter import ttk, messagebox
import psycopg2
DB_CONFIG = {
    "host": "localhost",
    "database": "practica",
    "user": "postgres",
    "password": "Nik12345",
    "port": "5432"
}
def get_user_by_login(login):
    """Возвращает (fio, role) или None."""
    conn = None
    try:
        conn = psycopg2.connect(**DB_CONFIG)
        with conn.cursor() as cur:
            cur.execute("""
                SELECT u.last_name, u.first_name, u.middle_name, r.roles_name
                FROM users u
                LEFT JOIN roles r ON u.roles_id = r.roles_id
                WHERE u.login = %s;
            """, (login,))
            row = cur.fetchone()
            if row:
                fio = f"{row[0] or ''} {row[1] or ''} {row[2] or ''}".strip()
                return {"fio": fio, "role": row[3] or "user"}
    except Exception as e:
        messagebox.showerror("Ошибка БД", str(e))
    finally:
        if conn:
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            conn.close()
    return None
def fetch_users():
    """Загружает пользователей в Treeview."""
    conn = None
    try:
        conn = psycopg2.connect(**DB_CONFIG)
        cur = conn.cursor()
        cur.execute("""
            SELECT u.users_id, u.first_name, u.last_name, u.middle_name,
                   u.login, r.roles_name
            FROM users u
            LEFT JOIN roles r ON u.roles_id = r.roles_id;
        """)
        rows = cur.fetchall()
        for item in tree.get_children():
            tree.delete(item)
        for row in rows:
            tree.insert('', 'end', values=row)
        cur.close()
    except Exception as e:
        messagebox.showerror("Ошибка БД", str(e))
    finally:
        if conn:
            conn.close()
def clear_root():
    for w in root.winfo_children():
        w.destroy()
def show_catalog_screen():
    clear_root()
    root.title("Каталог")
    root.geometry("1000x650")
    root.resizable(True, True)
    header = tk.Frame(root)
    header.pack(fill="x", pady=(20, 10))
    tk.Button(
        header, text="← Назад", font=("Arial", 11),
        command=lambda: show_main_screen(current_user)
    ).pack(side="left", padx=20)
    tk.Label(
        header, text="Магазин BURMOLDA",
        font=("Arial", 20, "bold")
    ).pack(side="left", expand=True)
    container = tk.Frame(root)
    container.pack(fill="both", expand=True, padx=20, pady=10)
    def create_card(parent, production, name, category, count, sostav, price):
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        card = tk.Frame(parent, highlightbackground="black",
                        highlightthickness=2, bd=0)
        card.pack(pady=10, padx=20, fill="x")
        photo = tk.Frame(card, width=100, height=100,
                         highlightbackground="black", highlightthickness=2)
        photo.pack(side="left", padx=15, pady=15)
        photo.pack_propagate(False)
        tk.Label(photo, text="ФОТО", font=("Arial", 14, "bold")).place(
            relx=0.5, rely=0.5, anchor="center")
        info = tk.Frame(card)
        info.pack(side="left", fill="both", expand=True, padx=10, pady=10)
        top_row = tk.Frame(info)
        top_row.pack(anchor="w", pady=(0, 10))
        tk.Label(top_row, text=production, font=("Arial", 14,
"bold")).pack(side="left")
        tk.Label(top_row, text=" | ", font=("Arial", 14,
"bold")).pack(side="left")
        tk.Label(top_row, text=name, font=("Arial", 14,
"bold")).pack(side="left")
        tk.Label(info, text=f"Категория: {category}",
                 font=("Arial", 13), anchor="w").pack(fill="x", pady=2)
        tk.Label(info, text=f"Количество: {count}",
                 font=("Arial", 13), anchor="w").pack(fill="x", pady=2)
        tk.Label(info, text=f"Состав: {sostav}",
                 font=("Arial", 13), anchor="w").pack(fill="x", pady=2)
        tk.Label(info, text=f"Цена: {price}",
                 font=("Arial", 14, "bold"), anchor="e").pack(fill="x", pady=
(10, 0))
    create_card(container, "Производство 1", "Товар 1", "Категория A", 10,
"...", "1000 ₽")
    create_card(container, "Производство 2", "Товар 2", "Категория B", 5,
"...", "2000 ₽")
def show_users_screen():
    clear_root()
    root.title("Пользователи")
    root.geometry("900x450")
    root.resizable(True, True)
    header = tk.Frame(root)
    header.pack(fill="x", pady=(10, 5))
    tk.Button(header, text="← Назад", font=("Arial", 11),
              command=lambda:
show_main_screen(current_user)).pack(side="left", padx=20)
    tk.Label(header, text="Список пользователей",
             font=("Arial", 14, "bold")).pack(side="left", expand=True)
    global tree
    columns = ("users_id", "first_name", "last_name", "middle_name", "login",
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"roles_name")
    tree = ttk.Treeview(root, columns=columns, show="headings")
    tree.heading("users_id", text="ID")
    tree.heading("first_name", text="Имя")
    tree.heading("last_name", text="Фамилия")
    tree.heading("middle_name", text="Отчество")
    tree.heading("login", text="Логин")
    tree.heading("roles_name", text="Роль")
    tree.column("users_id", width=60, anchor="center")
    tree.column("first_name", width=120)
    tree.column("last_name", width=120)
    tree.column("middle_name", width=120)
    tree.column("login", width=130)
    tree.column("roles_name", width=130)
    tree.pack(fill="both", expand=True, padx=10, pady=5)
    tk.Button(root, text="Обновить", command=fetch_users,
              font=("Arial", 12)).pack(pady=5)
    fetch_users()
current_user = None
def show_main_screen(user):
    global current_user
    current_user = user
    clear_root()
    root.title("Главная страница")
    root.geometry("600x450")
    root.resizable(True, True)
    tk.Label(
        root,
        text=f"Добро пожаловать,\n{user['fio']}!",
        font=("Arial", 18, "bold"),
        pady=20
    ).pack()
    tk.Label(
        root,
        text=f"Ваша роль: {user['role']}",
        font=("Arial", 12),
        fg="#555"
    ).pack(pady=(0, 20))
    btn_frame = tk.Frame(root)
    btn_frame.pack(pady=10)
    tk.Button(
        btn_frame, text="Каталог товаров",
        font=("Arial", 12), width=25, height=2,
        command=show_catalog_screen
    ).pack(pady=5)
    if user.get("role", "").lower() in ("admin", "администратор"):
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        tk.Button(
            btn_frame, text="Пользователи",
            font=("Arial", 12), width=25, height=2,
            command=show_users_screen
        ).pack(pady=5)
    tk.Button(
        root, text="Выйти",
        command=show_login_screen,
        bg="#f44336", fg="white",
        font=("Arial", 11), width=15
    ).pack(side="bottom", pady=20)
def show_login_screen():
    clear_root()
    root.title("Вход в систему")
    root.geometry("400x280")
    root.resizable(False, False)
    tk.Label(
        root, text="Магазин BURMOLDA",
        font=("Arial", 16, "bold")
    ).pack(pady=(20, 10))
    tk.Label(root, text="Логин пользователя",
             font=("Arial", 11)).pack(pady=(10, 5))
    entry_login = tk.Entry(root, width=25, font=("Arial", 12))
    entry_login.pack()
    entry_login.focus()
    def handle_login():
        login = entry_login.get().strip()
        if not login:
            messagebox.showwarning("Ошибка", "Введите логин!")
            return
        user = get_user_by_login(login)
        if user:
            show_main_screen(user)
        else:
            messagebox.showerror("Ошибка", "Пользователь не найден!")
    entry_login.bind('<Return>', lambda e: handle_login())
    tk.Button(
        root, text="Войти в систему",
        font=("Arial", 11), width=20,
        command=handle_login,
        relief="solid", borderwidth=2
    ).pack(pady=20)
if __name__ == "__main__":
    root = tk.Tk()
    show_login_screen()
    root.mainloop()
```